

New York TLC Project | Regression Model

Executive Summary



Automatidata

Overview:

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To predict fare amount on New York City TLC app, this summary shows results of the work done by our team to build a regression model for ride fares based on a variety of variables.

Problem

EDA showed that means of trip duration and distance have the most impact on fare amount, but these two variables don't obey the assumption of no multicollinearity.

Solution

Since the model's purpose is to use it for prediction, not discovering statistical inferences in the data. Multivariable linear regression based on these two variables will be appropriate.

Details

- EDA were conducted that included imputing outliers in variables of interest and engineering new data features. These are mean duration and mean distance of the trip. And these variables were grouped based on pickup and dropoff locations.
- The model developed showed that up to 87% of the variance in the dependent variables can be explained by these two variables.
- The model predicts that an increase of one standard deviation in mean distance make the fare amount tend to increase by 7.12 dollars. Similarly for mean duration where an increase of one standard deviation in the mean make fare amount tend to increase by 2.8 dollars.
- The model assumptions were then validated.
- A note that might be of interest is that some records are of no need to be predicted which are flat fare like JFK airport with a fare amount of 52\$.

Next Steps

- The model can be improved by adding more variables such as rush hours and passengers count. Then its predicted values can be substituted to machine learning models through pipelines.